

LSTM Backpropagation Derivatives (Fully Expanded Form)

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For simplicity and clarity, the following shorthand is used in the derivatives:

- **Let** $\sigma'(z) = \sigma(z) \odot (1 - \sigma(z))$ (sigmoid derivative)
- **Let** $\tanh'(z) = 1 - \tanh^2(z)$ (tanh derivative)

Common Building Blocks (Fully Expanded)

Key intermediate gradients:

$$\begin{aligned}\frac{\partial h_t}{\partial o_t} &= \tanh(C_t) \\ \frac{\partial h_t}{\partial C_t} &= o_t \odot (1 - \tanh^2(C_t)) \\ \frac{\partial h_t}{\partial f_t} &= \frac{\partial h_t}{\partial C_t} \odot C_{t-1} \\ \frac{\partial h_t}{\partial i_t} &= \frac{\partial h_t}{\partial C_t} \odot \tilde{C}_t \\ \frac{\partial h_t}{\partial \tilde{C}_t} &= \frac{\partial h_t}{\partial C_t} \odot i_t\end{aligned}$$

Where:

- $o_t = \sigma(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t})$
- $f_t = \sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t})$
- $i_t = \sigma(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t})$
- $\tilde{C}_t = \tanh(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t})$
- $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$

$$\begin{aligned}h_t &= \sigma(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \odot \tanh(\sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \odot C_{t-1}) + \\ &\quad \sigma(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \odot \tanh(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t})\end{aligned}$$

$$C_t = \sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \odot C_{t-1} + \sigma(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \odot \tanh(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t}) \quad (1)$$

1. Derivatives with respect to Output Gate Parameters

$$\begin{aligned}\frac{\partial h_t}{\partial w_{ho_t}} &= \left(\frac{\partial h_t}{\partial o_t} \odot \sigma'(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \right) \cdot h_{t-1}^\top \\ \frac{\partial h_t}{\partial w_{xo_t}} &= \left(\frac{\partial h_t}{\partial o_t} \odot \sigma'(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \right) \cdot x_t^\top \\ \frac{\partial h_t}{\partial b_{o_t}} &= \frac{\partial h_t}{\partial o_t} \odot \sigma'(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t})\end{aligned}$$

2. Derivatives with respect to Forget Gate Parameters

$$\begin{aligned}\frac{\partial h_t}{\partial w_{hf_t}} &= \left(\frac{\partial h_t}{\partial f_t} \odot \sigma'(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \right) \cdot h_{t-1}^\top \\ \frac{\partial h_t}{\partial w_{xf_t}} &= \left(\frac{\partial h_t}{\partial f_t} \odot \sigma'(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \right) \cdot x_t^\top \\ \frac{\partial h_t}{\partial b_{f_t}} &= \frac{\partial h_t}{\partial f_t} \odot \sigma'(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t})\end{aligned}$$

3. Derivatives with respect to Input Gate Sigmoid Parameters

$$\begin{aligned}\frac{\partial h_t}{\partial w_{hi_{1t}}} &= \left(\frac{\partial h_t}{\partial i_t} \odot \sigma'(w_{hi_{1t}} \cdot h_{t-1} + w_{xi_{1t}} \cdot x_t + b_{i_{1t}}) \right) \cdot h_{t-1}^\top \\ \frac{\partial h_t}{\partial w_{xi_{1t}}} &= \left(\frac{\partial h_t}{\partial i_t} \odot \sigma'(w_{hi_{1t}} \cdot h_{t-1} + w_{xi_{1t}} \cdot x_t + b_{i_{1t}}) \right) \cdot x_t^\top \\ \frac{\partial h_t}{\partial b_{i_{1t}}} &= \frac{\partial h_t}{\partial i_t} \odot \sigma'(w_{hi_{1t}} \cdot h_{t-1} + w_{xi_{1t}} \cdot x_t + b_{i_{1t}})\end{aligned}$$

4. Derivatives with respect to Candidate Parameters

$$\begin{aligned}\frac{\partial h_t}{\partial w_{hi_{2t}}} &= \left(\frac{\partial h_t}{\partial \tilde{C}_t} \odot (1 - \tanh^2(w_{hi_{2t}} \cdot h_{t-1} + w_{xi_{2t}} \cdot x_t + b_{i_{2t}})) \right) \cdot h_{t-1}^\top \\ \frac{\partial h_t}{\partial w_{xi_{2t}}} &= \left(\frac{\partial h_t}{\partial \tilde{C}_t} \odot (1 - \tanh^2(w_{hi_{2t}} \cdot h_{t-1} + w_{xi_{2t}} \cdot x_t + b_{i_{2t}})) \right) \cdot x_t^\top \\ \frac{\partial h_t}{\partial b_{i_{2t}}} &= \frac{\partial h_t}{\partial \tilde{C}_t} \odot (1 - \tanh^2(w_{hi_{2t}} \cdot h_{t-1} + w_{xi_{2t}} \cdot x_t + b_{i_{2t}}))\end{aligned}$$

5. Derivatives with respect to Previous States and Input

$$\begin{aligned}
\frac{\partial h_t}{\partial h_{t-1}} &= \left(\frac{\partial h_t}{\partial o_t} \odot \sigma'(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \right) \cdot w_{ho_t} \\
&+ \left(\frac{\partial h_t}{\partial f_t} \odot \sigma'(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \right) \cdot w_{hf_t} \\
&+ \left(\frac{\partial h_t}{\partial i_t} \odot \sigma'(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \right) \cdot w_{hi1_t} \\
&+ \left(\frac{\partial h_t}{\partial \tilde{C}_t} \odot (1 - \tanh^2(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t})) \right) \cdot w_{hi2_t}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h_t}{\partial C_{t-1}} &= \frac{\partial h_t}{\partial C_t} \odot f_t \\
&= o_t \odot (1 - \tanh^2(C_t)) \odot \sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t})
\end{aligned}$$

$$\begin{aligned}
\frac{\partial h_t}{\partial x_t} &= \left(\frac{\partial h_t}{\partial o_t} \odot \sigma'(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \right) \cdot w_{xo_t} \\
&+ \left(\frac{\partial h_t}{\partial f_t} \odot \sigma'(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \right) \cdot w_{xf_t} \\
&+ \left(\frac{\partial h_t}{\partial i_t} \odot \sigma'(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \right) \cdot w_{xi1_t} \\
&+ \left(\frac{\partial h_t}{\partial \tilde{C}_t} \odot (1 - \tanh^2(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t})) \right) \cdot w_{xi2_t}
\end{aligned}$$

These are all the derivatives with respect to every variable appearing in the expanded equation for h_t .